

Explanation – HW12: Diagonalization & Applications

Part 1: Foundations – explained simply

1. What does “diagonalizing” mean? The right glasses

A matrix A deforms space – usually in a complicated way (rotating *and* stretching). But in HW11 we saw: there are special directions, the **eigenvectors**, that A only *stretches*. Diagonalizing means: look at space through the “eigenvector glasses”. Through these glasses A does nothing complicated anymore – it just stretches each axis by its eigenvalue. That is the **diagonal matrix** D .

$$\begin{array}{ccc} v \text{ (standard)} & \xrightarrow{A \text{ (complicated)}} & Av \text{ (standard)} \\ \downarrow P^{-1} & & \uparrow P \\ \text{eigenbasis} & \xrightarrow{D \text{ (just stretch)}} & \text{eigenbasis} \end{array}$$

Analogy

Imagine describing a landscape. In normal compass directions (north/east) a slope is “slanted” and awkward to compute. But if you turn so you look *straight up the slope*, everything is simple: ahead it only goes up, sideways it stays flat. The eigenvectors are these “convenient directions”, and D is the landscape in that view.

The formula

A is **diagonalizable** if

$$D = P^{-1}AP \quad \text{resp.} \quad A = PDP^{-1},$$

with P invertible and D diagonal. Here the **columns of P are the eigenvectors** and the **diagonal of D holds the corresponding eigenvalues** (same order!).

2. When does it work? Enough eigenvectors

For P to be invertible, its columns – the eigenvectors – must form a **whole basis**. For an $n \times n$ matrix you therefore need n linearly independent eigenvectors.

Diagonalizability criterion

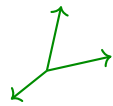
A ($n \times n$) is diagonalizable

$\iff$ there are n independent eigenvectors $\iff$ for *every* eigenvalue GM = AM.

As soon as *one* eigenvalue has GM < AM (an eigenvector “is missing”), A is **not** diagonalizable.

Analogy

For $\mathbb{R}^3$ you need exactly 3 “poles” pointing in different directions to build a scaffold (a basis). With only 2 poles they lie in a plane – the space stays incomplete. Likewise: 2 eigenvectors are not enough in $\mathbb{R}^3$.



3 directions = basis ✓
diagonalizable



only 2 directions = plane
not diagonalizable

3. What is it good for? Powers become easy

The biggest advantage: with $A = PDP^{-1}$ every **power** becomes child’s play. The middle $P^{-1}P$ cancel out:

$$A^k = (PDP^{-1})(PDP^{-1}) \cdots (PDP^{-1}) = P D^k P^{-1},$$

and D^k is just the diagonal to the power k :

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \Rightarrow D^k = \begin{pmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{pmatrix}.$$

Application: sequences

A recurrence like $a_k = 3a_{k-1} + 10a_{k-2}$ is written as $v_k = Mv_{k-1}$, i.e. $v_k = M^k v_0$. Via $M^k = PD^k P^{-1}$ you get an **explicit formula** for a_k – without computing all the intermediate terms.

4. Parameter problems: “for which a ?”

Sometimes a letter a sits in the matrix. Then the eigenvalues depend on a , and you ask: for which a is A diagonalizable? **Strategy:** determine the eigenvalues (with AM) as expressions in a , then check the GM of the repeated eigenvalue. Usually there are only *few* “bad” a where two eigenvalues collide and an eigenvector is missing.

Two useful structures (from HW11)

- $A = cI + (\mathbf{all-ones\ matrix})$: subtracting cI only shifts the eigenvalues. The all-ones matrix has rank 1 $\Rightarrow$ large eigenspace.
- **Rank-1 matrix** (all rows multiples of the same row): the eigenvalues are trace (once) and 0 (otherwise). $\text{GM}(0) = n - 1$.

Part 2: Overview of the exercises

Exercise 47 – diagonalize (with HW11 data)

We already know the three matrices from HW11 (43, 44). You start with the table (eigenvalues, AM, GM, bases) and only check GM = AM:

- (a) all GM = AM $\Rightarrow$ diagonalizable. P = eigenvectors as columns, $D = \text{diag}(3, 3, 5)$.
- (b) GM(2) = 1 < AM(2) = 2 $\Rightarrow$ only 2 instead of 3 eigenvectors $\Rightarrow$ *not* diagonalizable.
- (c) all GM = AM $\Rightarrow$ diagonalizable. $D = \text{diag}(0, 6, 6)$.

Instead of computing P^{-1} , you check $AP = PD$ – that means, column by column, just $Av_i = \lambda_i v_i$.

Exercise 48 – for which a ?

- (a) $A = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}$. The char. polynomial (Sarrus) gives eigenvalues $a - 1$ (double) and $a + 2$ (simple), *always* distinct. And $A - (a - 1)I$ is the all-ones matrix (rank 1) $\Rightarrow$ GM($a - 1$) = 2 = AM. So diagonalizable **for all** a .
- (b) $A = \begin{pmatrix} 1 & 2 & 3 \\ a & 2a & 3a \end{pmatrix}$ has rank 1 (all rows multiples of $(1, 2, 3)$). Eigenvalues 0 (AM 2, GM 2) and $5 + 3a$ (trace). As long as $5 + 3a \neq 0$ everything has GM = AM. At $a = -\frac{5}{3}$ the third eigenvalue is also 0 (AM 3), but GM stays 2 $\Rightarrow$ **only then not** diagonalizable.

Exercise 49 – sequence via diagonalization

- $v_k = (a_{k+1}, a_k)^\top$, then $v_k = Mv_{k-1}$ with $M = \begin{pmatrix} 3 & 10 \\ 1 & 0 \end{pmatrix}$, so $v_k = M^k v_0$, $v_0 = (-1, 4)^\top$.
- Diagonalize M : eigenvalues 5, -2, eigenvectors $(5, 1), (-2, 1)$.
- $v_k = PD^k P^{-1} v_0$. With $P^{-1} v_0 = (1, 3)^\top$ it follows that $a_k = 5^k + 3 \cdot (-2)^k$.

Summary

Exercise	Result
47 (a)	diagonalizable, $D = \text{diag}(3, 3, 5)$
47 (b)	<i>not</i> diagonalizable (GM(2) < AM(2))
47 (c)	diagonalizable, $D = \text{diag}(0, 6, 6)$
48 (a)	for all $a \in \mathbb{R}$
48 (b)	exactly for $a \neq -\frac{5}{3}$
49	$a_k = 5^k + 3 \cdot (-2)^k$

Golden rules

1. Diagonalizable $\Leftrightarrow$ for every eigenvalue GM = AM (enough eigenvectors for a basis).
2. P = eigenvectors as columns, D = eigenvalues on the diagonal – *same order*.
3. Check with $AP = PD$ (column by column $Av_i = \lambda_i v_i$) – no P^{-1} needed.
4. Powers: $A^k = PD^k P^{-1}$, and D^k is the diagonal to the power k .

5. Parameter: only look for the few a where $\text{GM} < \text{AM}$.
6. Sequences: pack into a vector, matrix M^k via diagonalization, read off the component.